Experiment 15

15.1.1 Car Dealership Sales System

Algorithm :

Step 1: Start
Step 2: Define base class Car
    2.1 Method __init__(self, brand, price, model, color)
        Assign self.brand = brand
        Assign self.price = price
        Assign self.model = model
        Assign self.color = color
    2.2 Method display_details(self)
        Print self.brand
        Print self.price
        Print self.model
        Print self.color
Step 3: Define derived class Car1 (inherits from Car)
    3.1 Method display_details(self)
        Call super().display_details()
        (Calls Car.display_details() of parent class)
Step 4: Define derived class Car2 (inherits from Car)
    4.1 Method display_details(self)
        Call super().display_details()
        (Calls Car.display_details() of parent class)
Step 5: Read input for car1
    5.1 Read a line of input and split into parts
    5.2 Assign brand1 = first part
    5.3 Assign price1 = float(second part)
    5.4 Assign model1 = third part
    5.5 Assign color1 = fourth part
Step 6: Read input for car2
    6.1 Read a line of input and split into parts
    6.2 Assign brand2 = first part
    6.3 Assign price2 = float(second part)
    6.4 Assign model2 = third part
    6.5 Assign color2 = fourth part
Step 7: Create objects
    7.1 car1 = Car1(brand1, price1, model1, color1)
        Calls Car.__init__() to initialize attributes
    7.2 car2 = Car2(brand2, price2, model2, color2)
        Calls Car.__init__() to initialize attributes
Step 8: Display details of car1
    8.1 Call car1.display_details()
    8.2 Car1.display_details() calls super().display_details()
    8.3 Car.display_details() executes:
        Print brand1
        Print price1
        Print model1
        Print color1

Step 9: Display details of car2

9.1 Call car2.display_details()

9.2 Car2.display_details() calls super().display_details()

9.3 Car.display_details() executes:

Print brand2

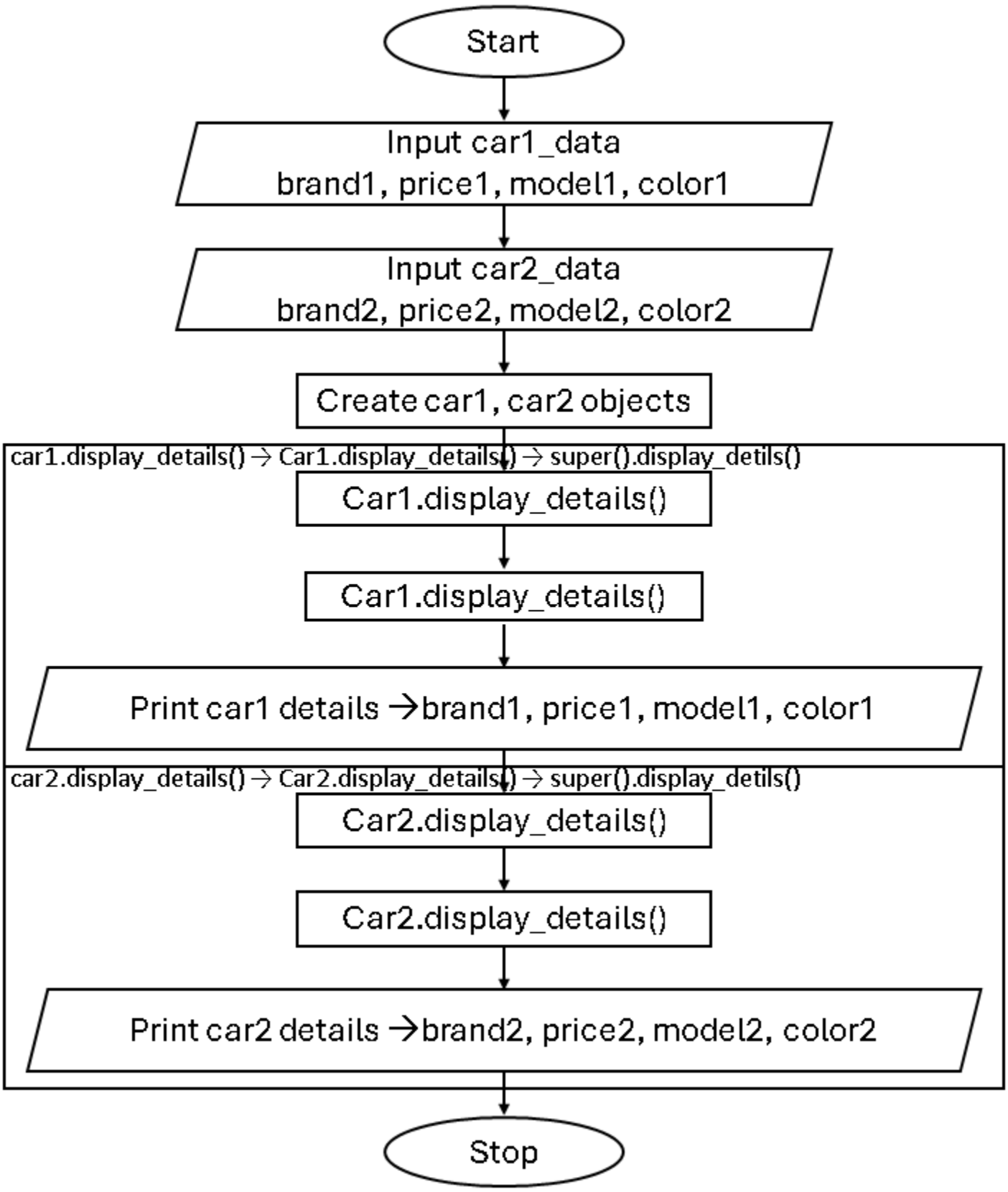
Print price2

Print model2

Print color2

Step 10: Stop

Flowchart :



Code :

```
class Car:
def __init__(self, brand, price, model,
color):
self.brand = brand
self.price = price
self.model = model
self.color = color


def display_details(self):
print(self.brand)
print(self.price)
print(self.model)
print(self.color)


class Car1(Car):
def display_details(self):
super().display_details()


class Car2(Car):
def display_details(self):
super().display_details()
# Read input
car1_data = input().split()
brand1, price1, model1, color1 =
car1_data[0], float(car1_data[1]),
car1_data[2], car1_data[3]


car2_data = input().split()
brand2, price2, model2, color2 =
car2_data[0], float(car2_data[1]),
car2_data[2], car2_data[3]
```

```
# Create objects
car1 = Car1(brand1, price1, model1, color1)
car2 = Car2(brand2, price2, model2, color2)

# Display details
car1.display_details()
car2.display_details()
```

Execution :